RAVNEET KAUR

Montreal | ravneetkaur24191508@gmail.com | linkedin.com/in/ravneet-kaur-bains/ | github.com/RAVNEETKAUR15

EDUCATION

M.Sc. Physics - McGill University, Montreal (3.9/4.0) **Jan 2024 - Aug 2026**

B.Sc. Physics (Honors) - SGTB Khalsa College, University of Delhi **July 2020 - May 2023**

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, Mathematica
Data & ML: pandas, NumPy, SciPy, Matplotlib, scikit-learn, LightGBM, XGBoost, DuckDB; feature engineering, time-series forecasting, walk-forward backtesting, model validation
AI/LLM: PyTorch, Hugging Face Transformers, LLM APIs (OpenAI/Anthropic), prompt engineering, AI-assisted development (Claude Code, GitHub Copilot)
Deployment & Computing: FastAPI, Streamlit, Docker, CI/CD, Git, Linux, Slurm, MPI, FFTW, parallel scaling, High-Performance computers (HPC)
Soft Skills: Communication, Problem Solving, Teamwork, Leadership, Time Management, Adaptability
Spoken Languages: English, French (A2-B1), Hindi, Punjabi

EXPERIENCE

Research Assistant - **Department of Physics, McGill University | Dr. Nikolas Provatas** **Sep 2026 – Oct 2026**

- Translated experimental 3D-printing temperature and time data from experimentalists into numerical simulation inputs by deriving length and time scaling factors to reproduce real thermal cycles.
- Ran 10-layer metal 3D-printing simulations on 2048x1024x1024 grids, producing 100TB+ of data on national HPC clusters by orchestrating Slurm job arrays.
- Developed a Python pipeline that turns raw 3D fields into per-atom datasets, using KD-tree neighbor search to separate matrix atoms from precipitate atoms.

Mitacs Accelerate Fellow – **McGill University (Industry partner: Hydro-Québec)** **Jan 2024 - Aug 2026**

- Extended **OpenPFC**, an open-source C++/MPI simulation framework, from single elements to binary alloys, and benchmarked parallel scaling up to 1,500+ cores.
- Implemented a semi-implicit FFT time-stepping scheme, allowing large, stable timesteps on 2-billion-point grids.
- Diagnosed and fixed a 47x scaling error between experimental input and simulation units, restoring physically consistent results that validated the model.

Internship - Student Consultant at PhonIA **June 2026**

- Showed that the 95-99% accuracy claims in stuttering-detection research came from speaker leakage by reviewing 7 published models against a realistic F1~0.65, and recommended speaker-independent evaluation.
- Found that 0 of 9 speech corpora were usable for French clinical training by auditing licensing and annotations, then scoped a data-collection plan covering 8 dysfluency classes.
- Proposed 5 upgrades to the prototype's Wav2Vec2 pipeline, including multi-task learning, audio augmentation and explainability, and presented an 18-month clinical validation roadmap to the CEO.

Teaching Assistant - Department of Physics, McGill University **Jan 2024 - Apr 2026**

- Taught quantitative problem-solving and experimental error analysis to 50+ undergraduates per term across five terms, by leading tutorials and laboratory sessions in introductory physics.

DATA SCIENCE PROJECTS

Horizon - Electricity Demand Forecasting - Python, LightGBM, DuckDB, SQL, pandas **Sept 2026**

- Built an end-to-end pipeline ingesting 8 years of hourly electricity data across **8 US grid regions** (**1M+** rows), with a data-quality audit catching overflow sentinels, feed dropouts and time-zone duplicates before training.
- Designed a walk-forward backtest scoring each 1-24h horizon independently, exposing signal decay with lead time instead of averaging it away.

IPL Chase Strategy Model - 1st place, ORU IPL Analytics Hackathon **April 2026**

- Converted 278,205 ball-by-ball records across 18 seasons into a live Streamlit win-probability tool.
- Raised chase-outcome accuracy from 76.1% baseline to **78.5%** via 10 match-state features and **108**-config GridSearchCV.
- Quantified a phase-dependent aggression trade-off: **+3.5%** win probability in death overs, **-10.9%** in the powerplay.

AWARDS & ACHIEVEMENTS

- Placed 2nd out of 47 in poster presentation at Grande Conférence du RQMP held at Université de Sherbrooke **June 2026**
- Awarded Certificate of Academic Achievement, ranked 2nd in academic session **2020-2021**